DATA ANALYST INTERNSHIP

Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset

Objective: Extract insights using visual and statistical exploration.

Tools: Python • Pandas • Matplotlib • Seaborn

Dataset: 891 rows × 12 original columns

1. Objective, Tools and Dataset

Objective

Extract insights using visual and statistical exploration, identifying patterns, relationships, trends and anomalies.

Tools

Python, Pandas, Matplotlib and Seaborn.

Dataset

The supplied Titanic training dataset contains 891 passenger records and 12 original columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin and Embarked.

EDA coverage

Dataset overview, data quality, descriptive statistics, univariate and bivariate analysis, feature engineering, multivariate analysis, correlation analysis, outliers, skewness, key findings and interview preparation.

2. Dataset Overview and Data Quality

Overview

Rows: 891

Columns: 12 original columns

Duplicate rows: 0

Overall survival rate: 38.38%

Missing values

Age: 177 missing (19.87%)

Cabin: 687 missing (77.10%)

Embarked: 2 missing (0.22%)

Interpretation

Age has substantial missingness, Cabin has very high missingness, and Embarked has only a small number of missing records. These issues should be considered before modeling.

Duplicates

No duplicate rows were found.

3. Statistical Summary

describe()

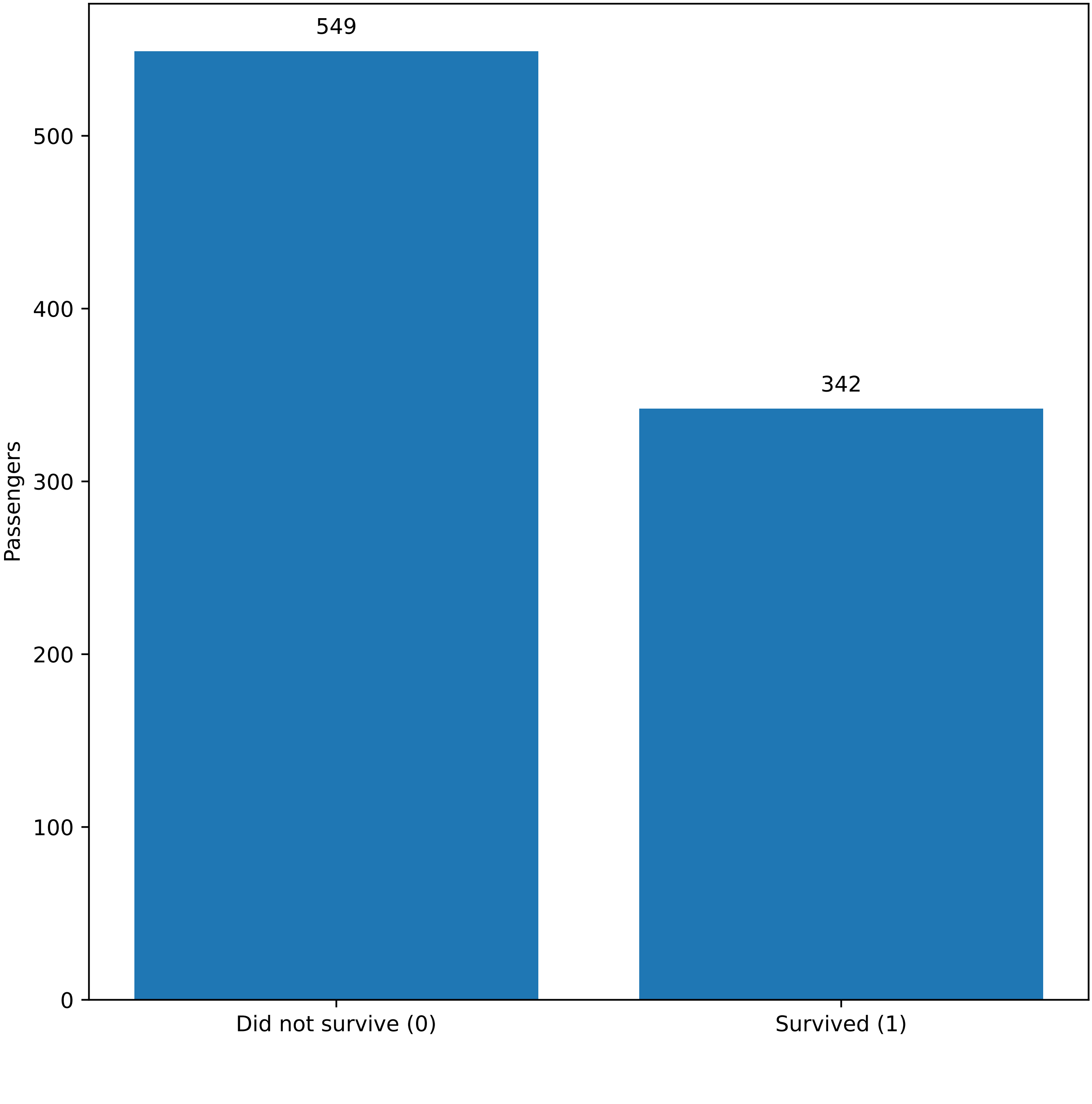
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	FamilySize
count	891.00	891.00	891.00	714.00	891.00	891.00	891.00	891.00
mean	446.00	0.38	2.31	29.70	0.52	0.38	32.20	1.90
std	257.35	0.49	0.84	14.53	1.10	0.81	49.69	1.61
min	1.00	0.00	1.00	0.42	0.00	0.00	0.00	1.00
25%	223.50	0.00	2.00	20.12	0.00	0.00	7.91	1.00
50%	446.00	0.00	3.00	28.00	0.00	0.00	14.45	1.00
75%	668.50	1.00	3.00	38.00	1.00	0.00	31.00	2.00
max	891.00	1.00	3.00	80.00	8.00	6.00	512.33	11.00

Interpretation

Fare has a wide range and a strongly skewed distribution. Age covers a broad range.

FamilySize was derived as SibSp + Parch + 1.

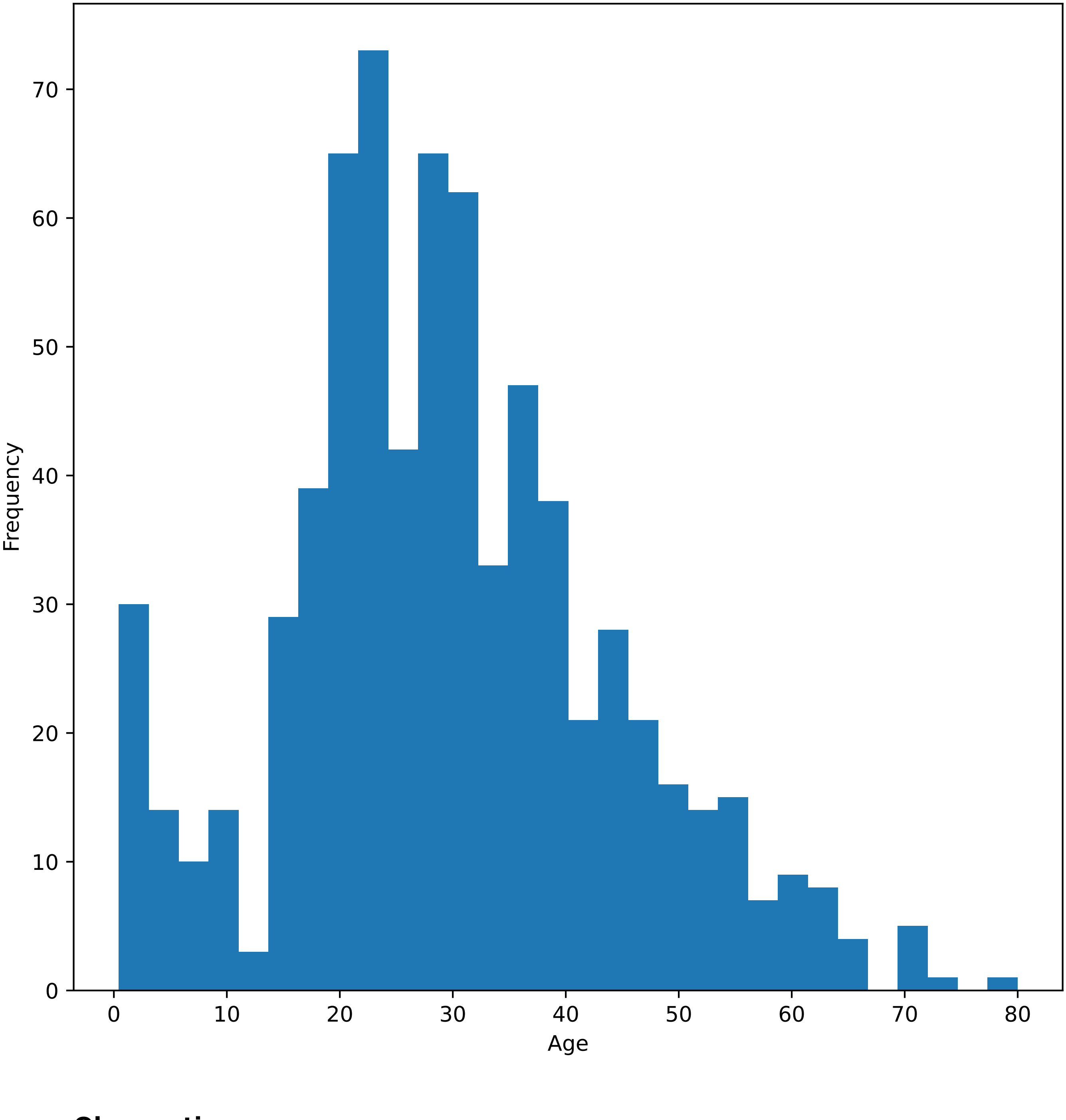
4. Univariate Analysis — Survival



Observation

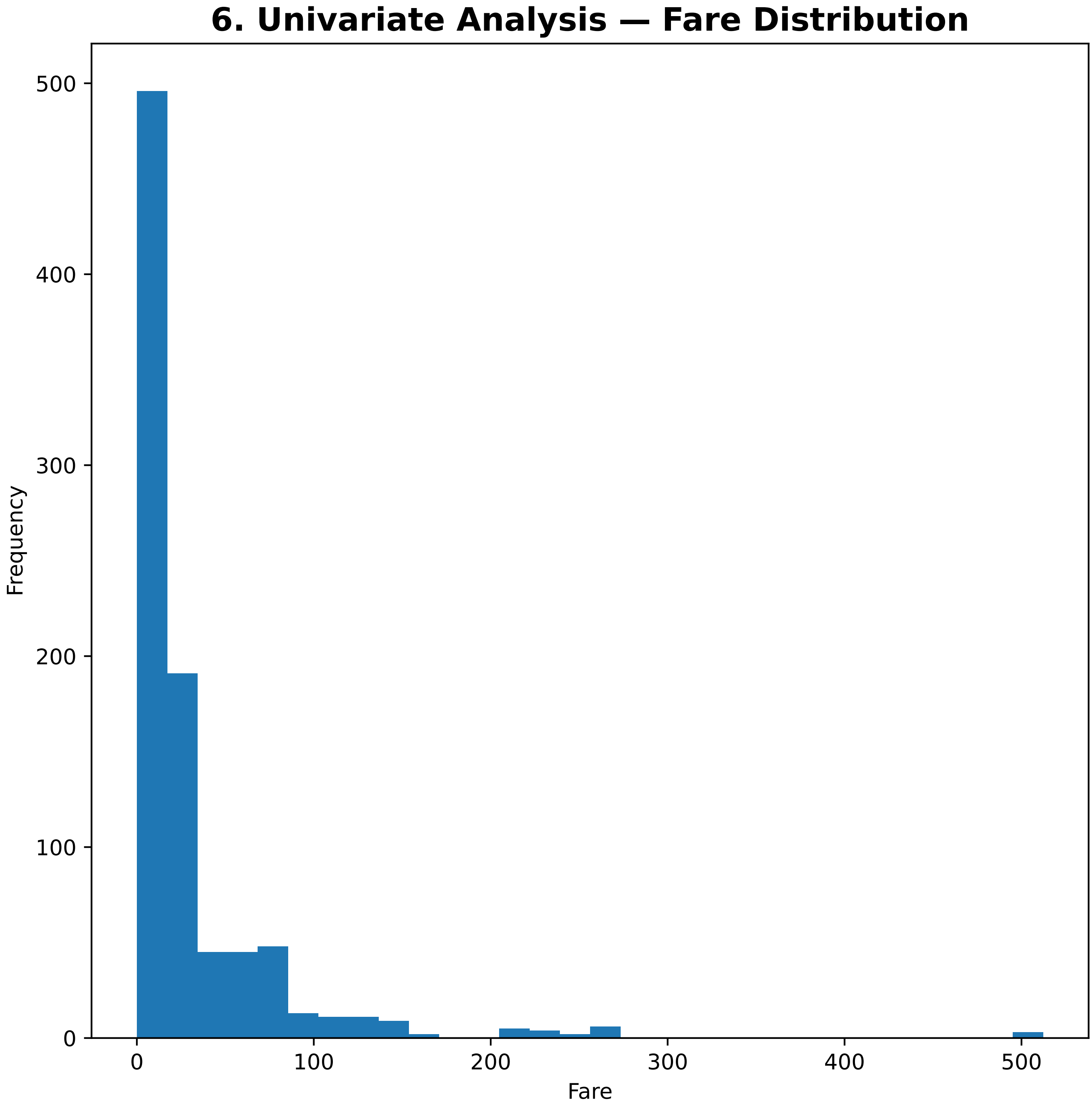
38.4% of passengers survived, so non-survivors form the larger group.

5. Univariate Analysis — Age Distribution



Observation

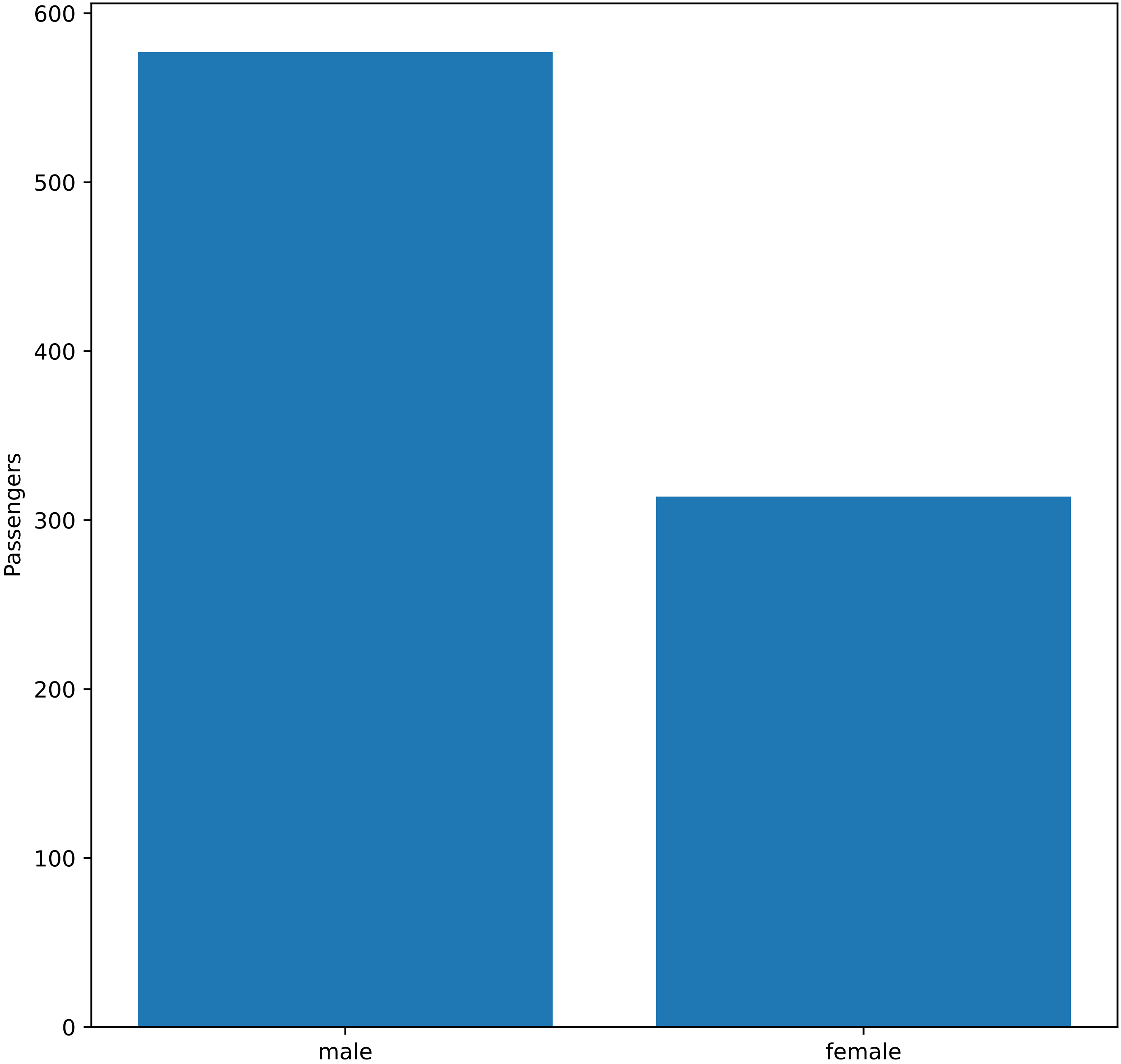
The available ages are concentrated around younger and middle-aged passengers. There are 177 missing Age values.



Observation

Fare is strongly right-skewed: most fares are relatively low, with a small number of very high fares.

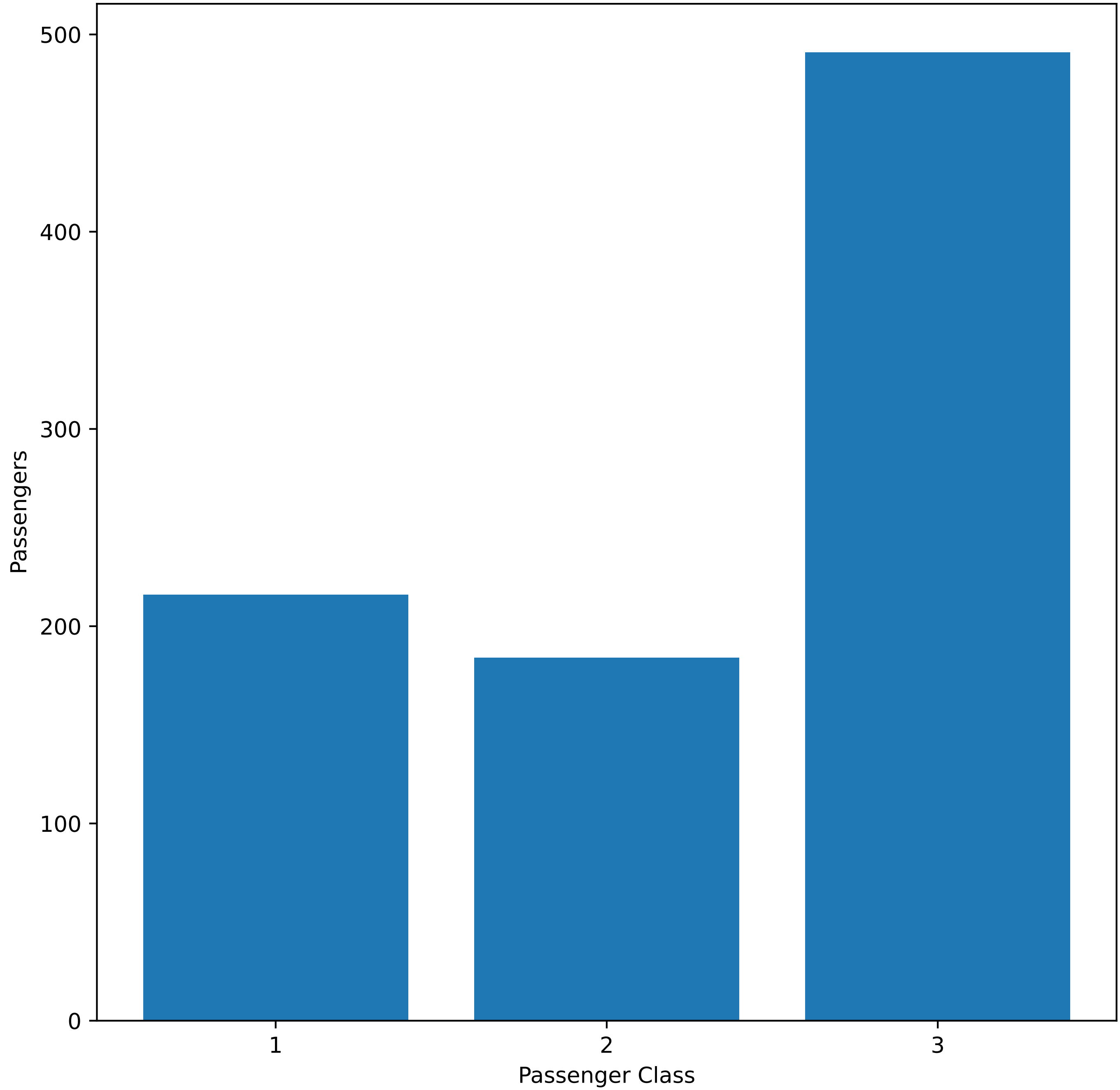
7. Univariate Analysis — Sex



Observation

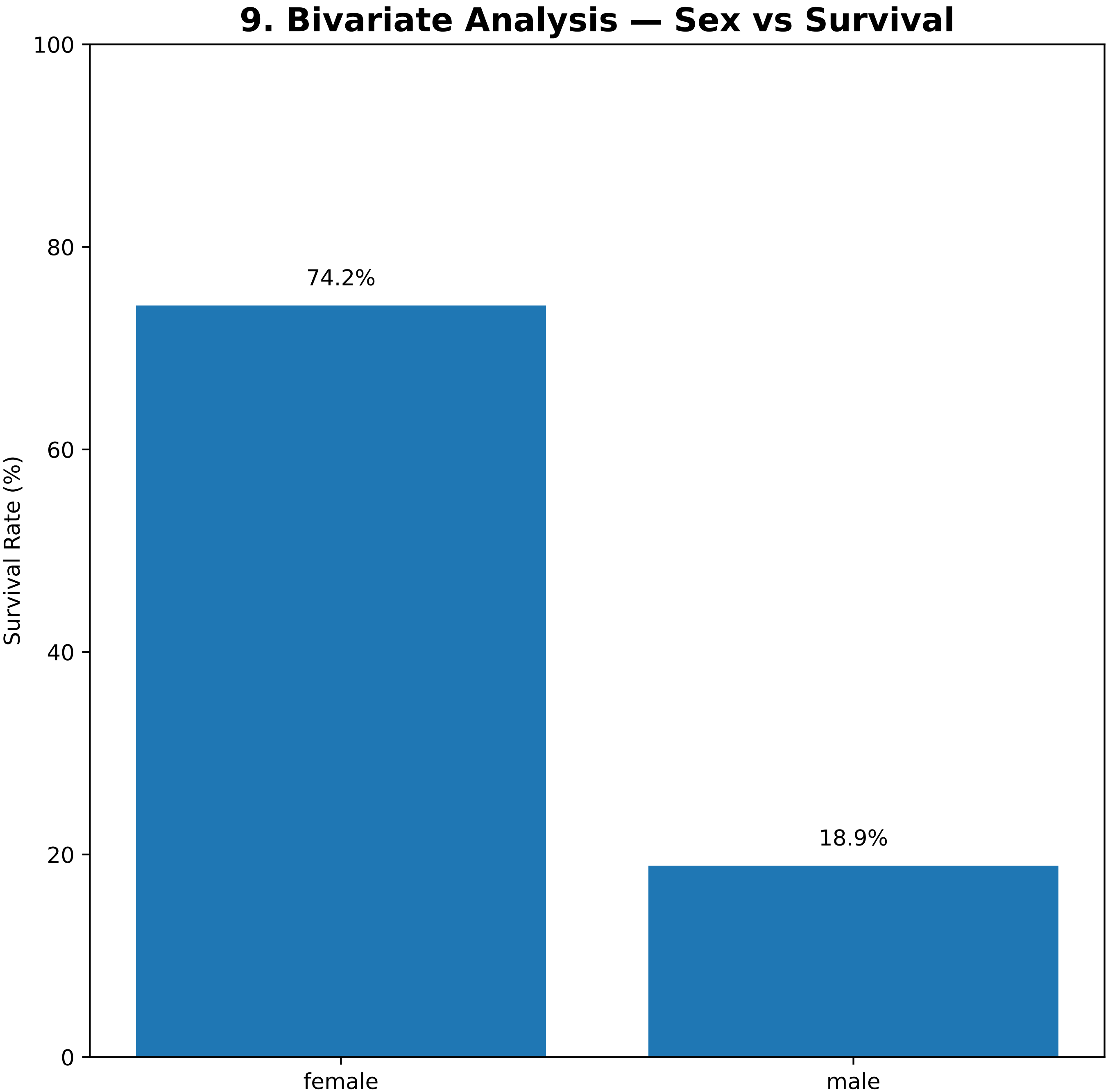
There are more male passengers than female passengers in the dataset.

8. Univariate Analysis — Passenger Class



Observation

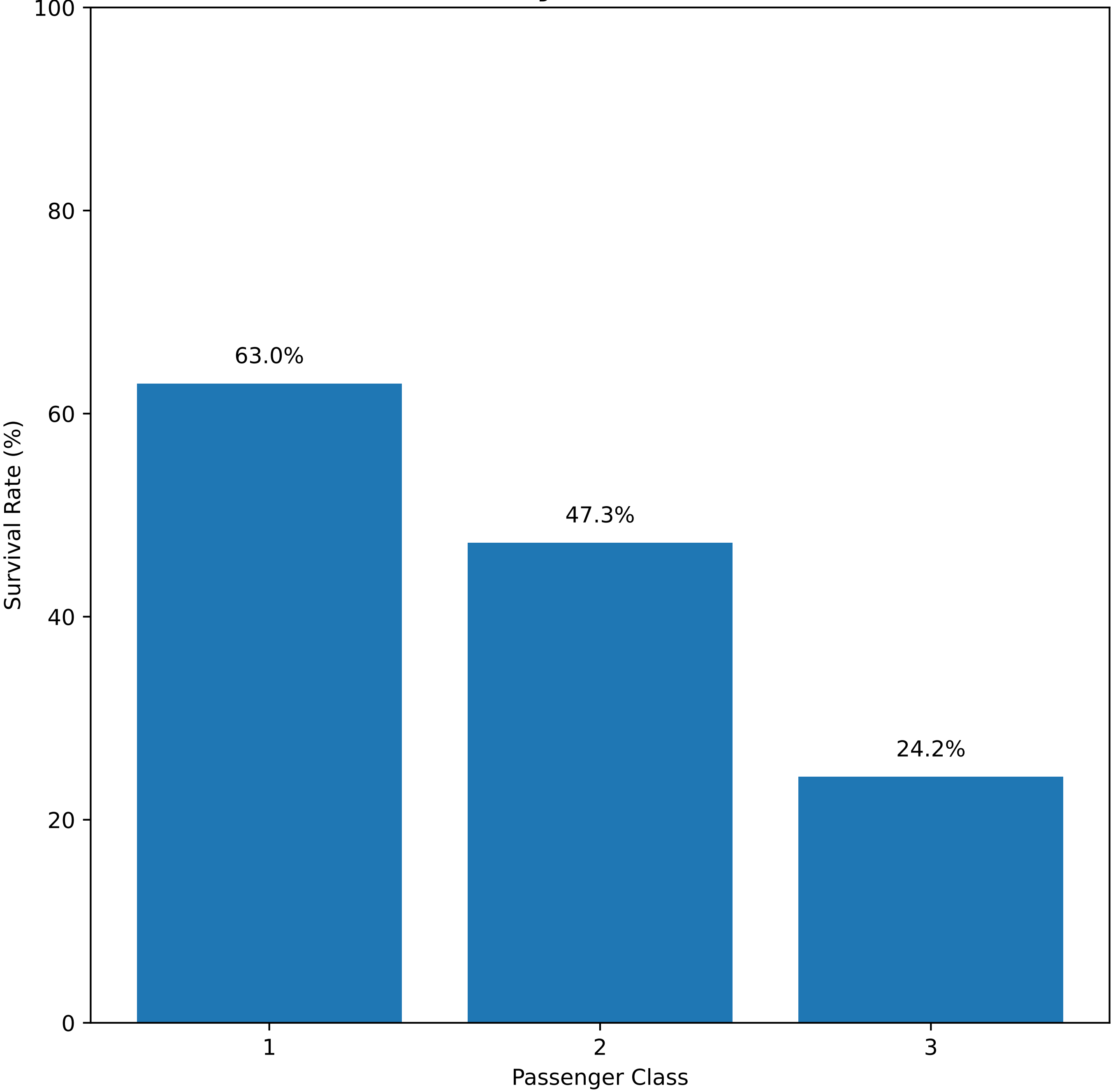
Third class contains the largest number of passengers, followed by first and second class.



Observation

Female survival was 74.2% versus 18.9% for males, showing a strong association between sex and survival.

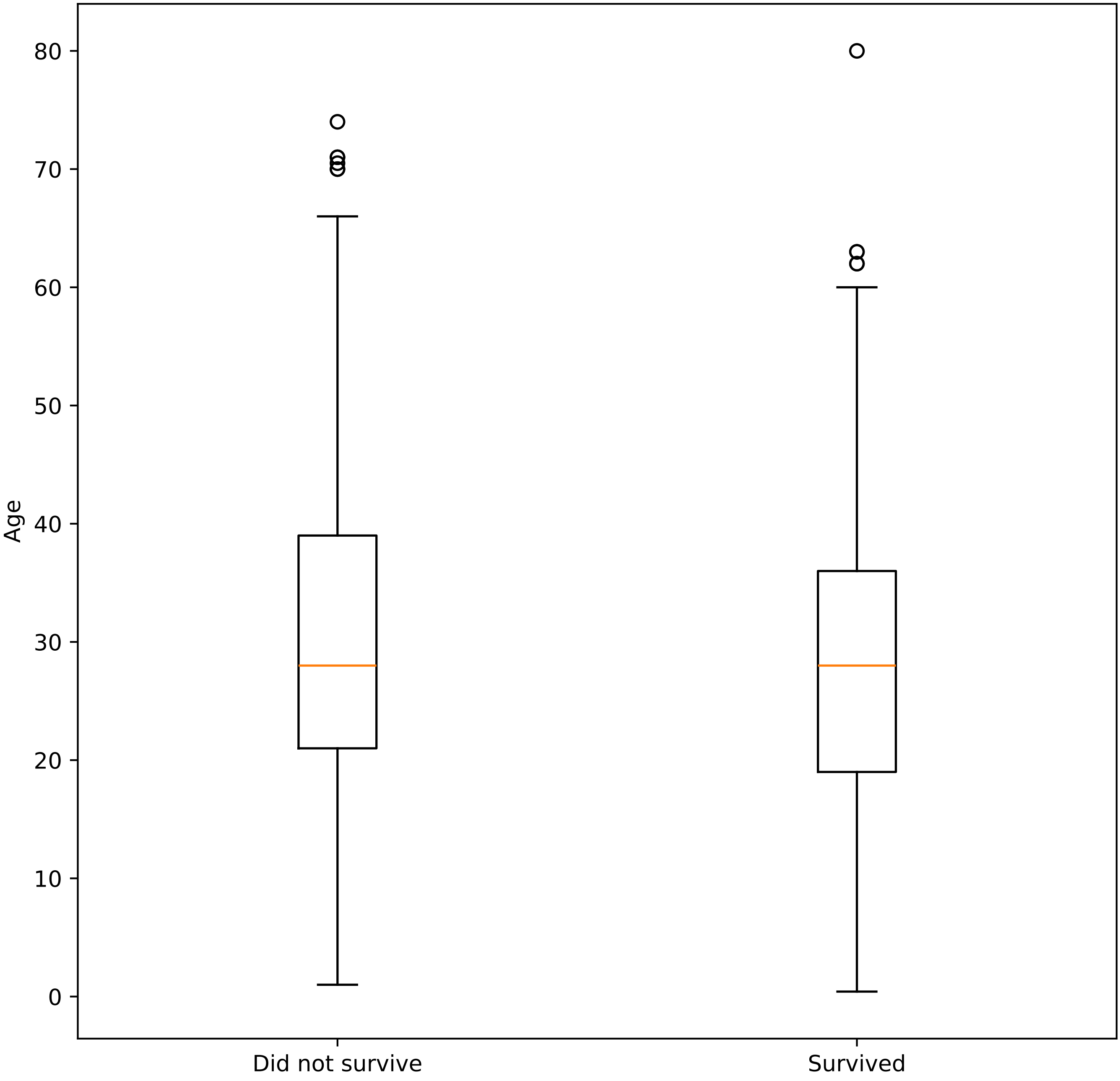
10. Bivariate Analysis — Pclass vs Survival



Observation

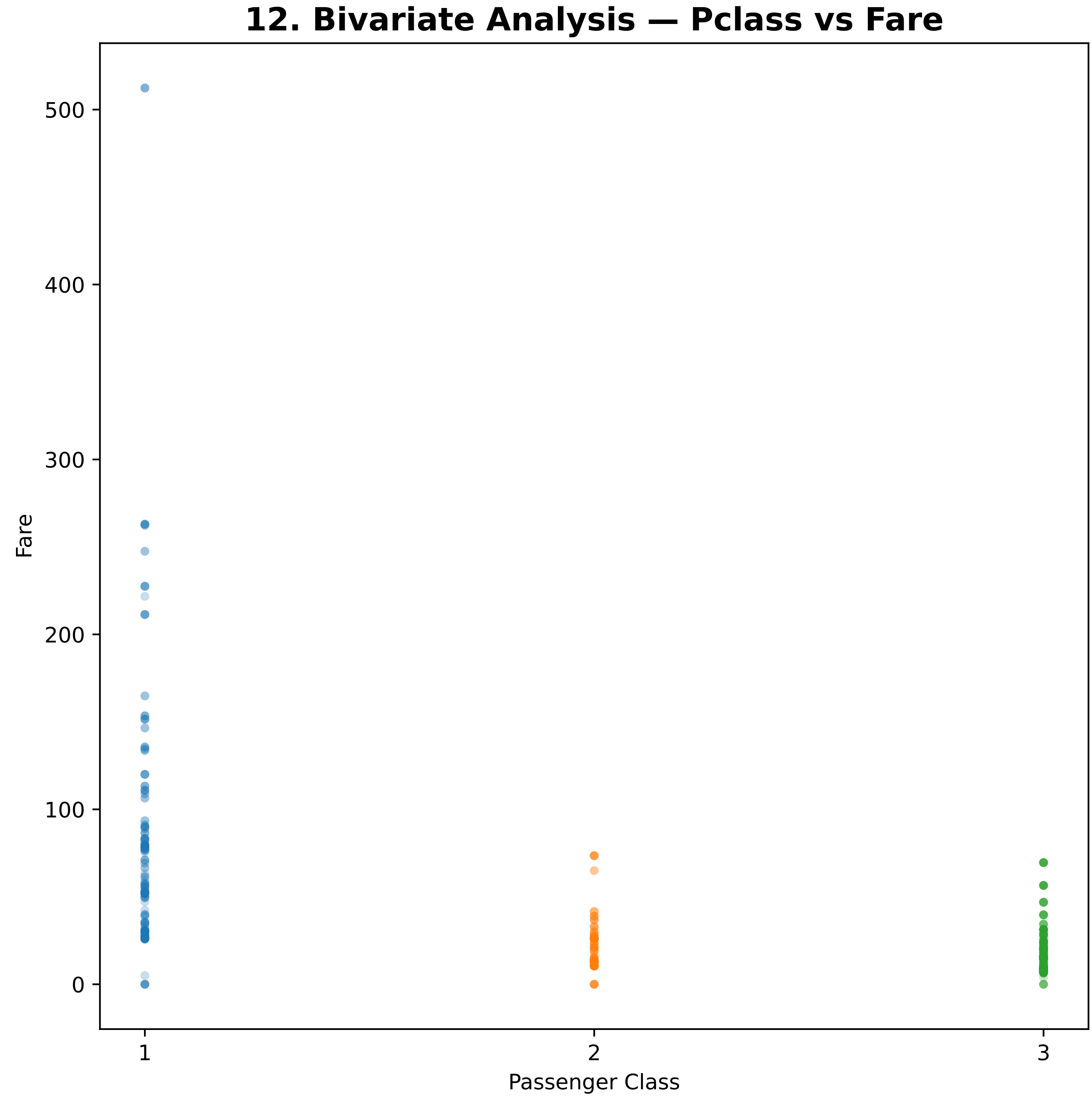
Survival decreases across classes: first class 63.0%, second class 47.3%, third class 24.2%.

11. Bivariate Analysis — Age vs Survival



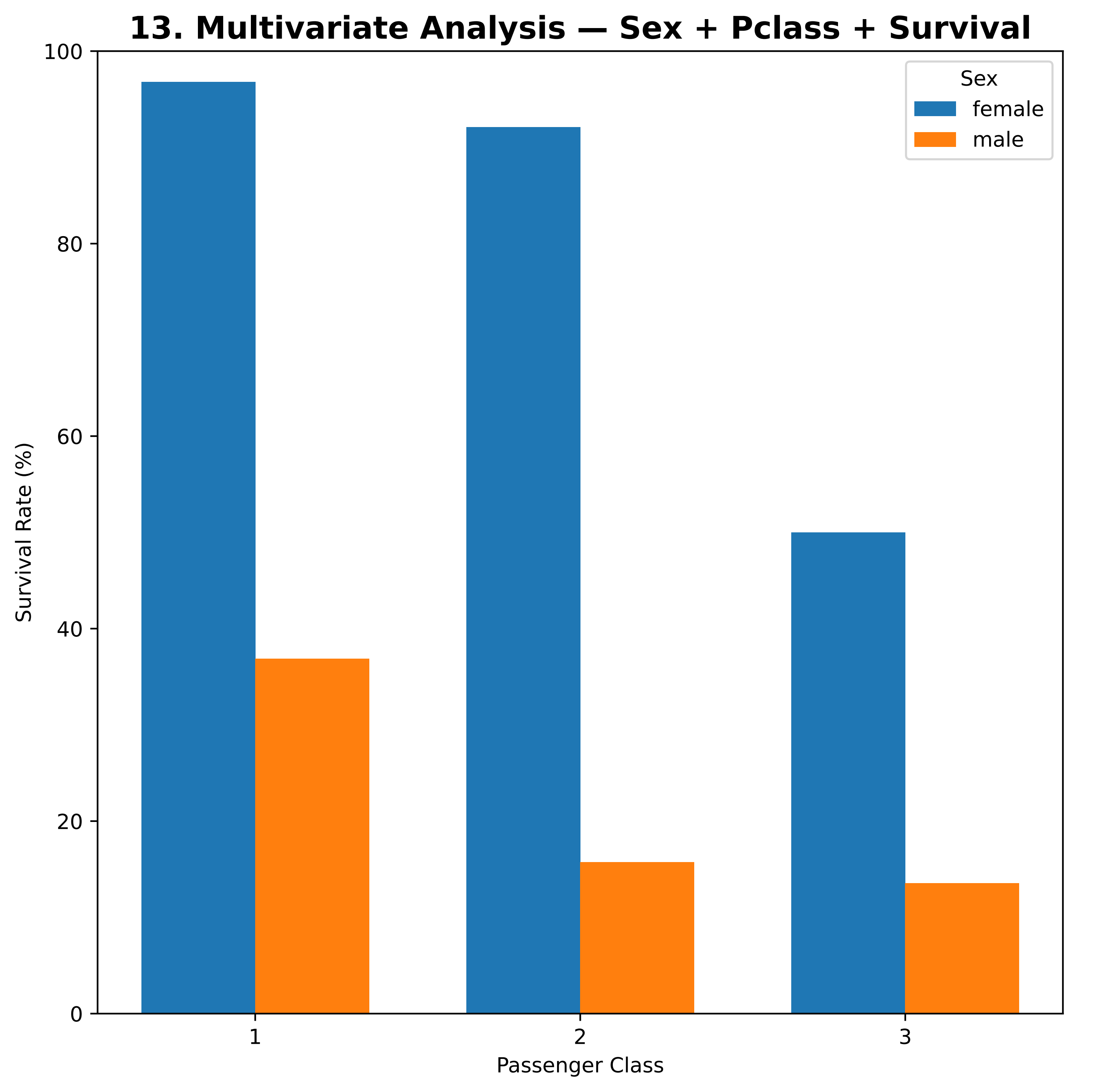
Observation

The age distributions overlap considerably, so age alone does not separate survivors from non-survivors as strongly as sex or passenger class.



Observation

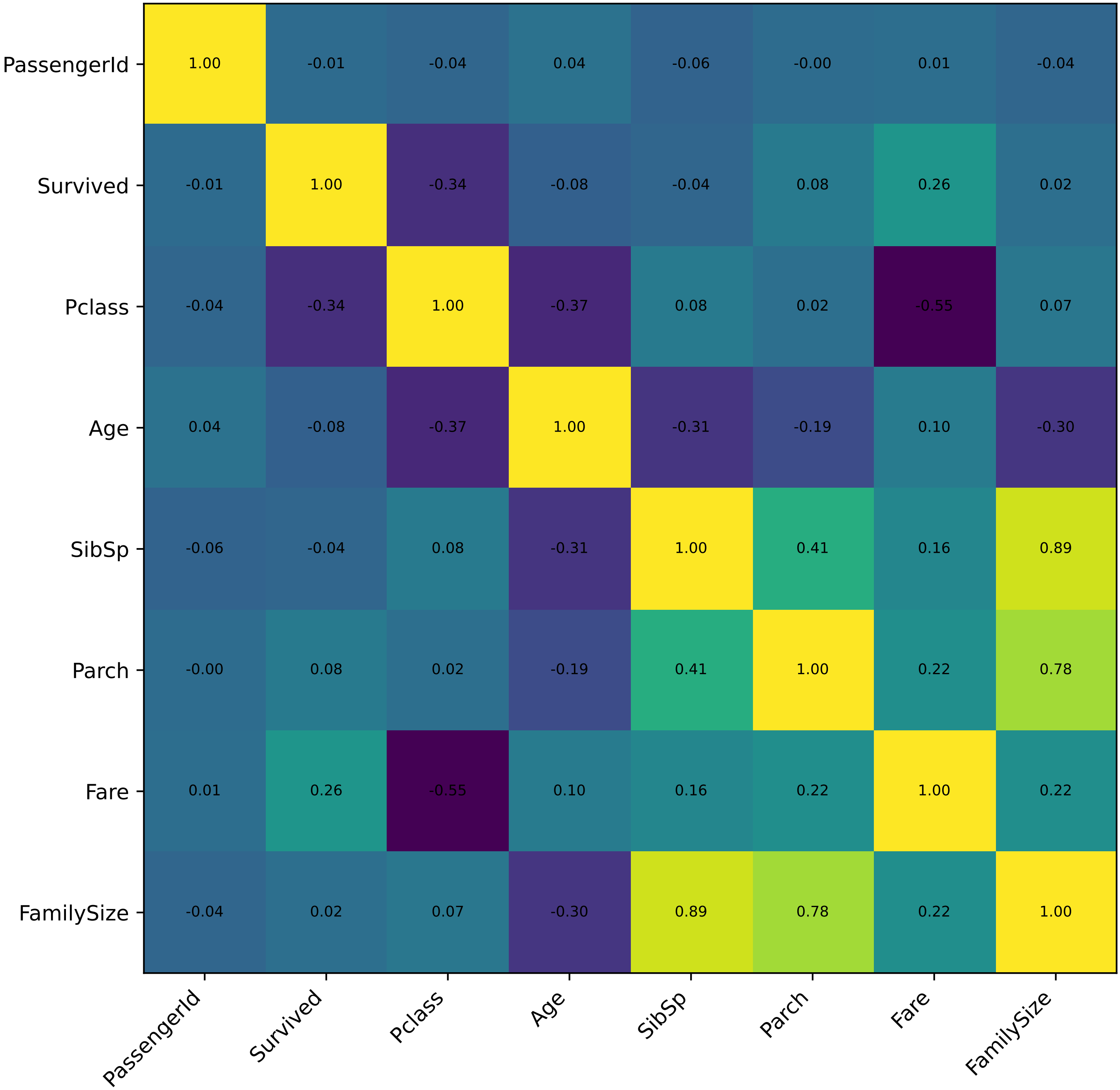
Higher passenger classes generally paid higher fares. First class also contains some particularly high-fare observations.



Observation

Female passengers generally had higher survival rates than males within the same class, while higher passenger classes generally had higher survival.

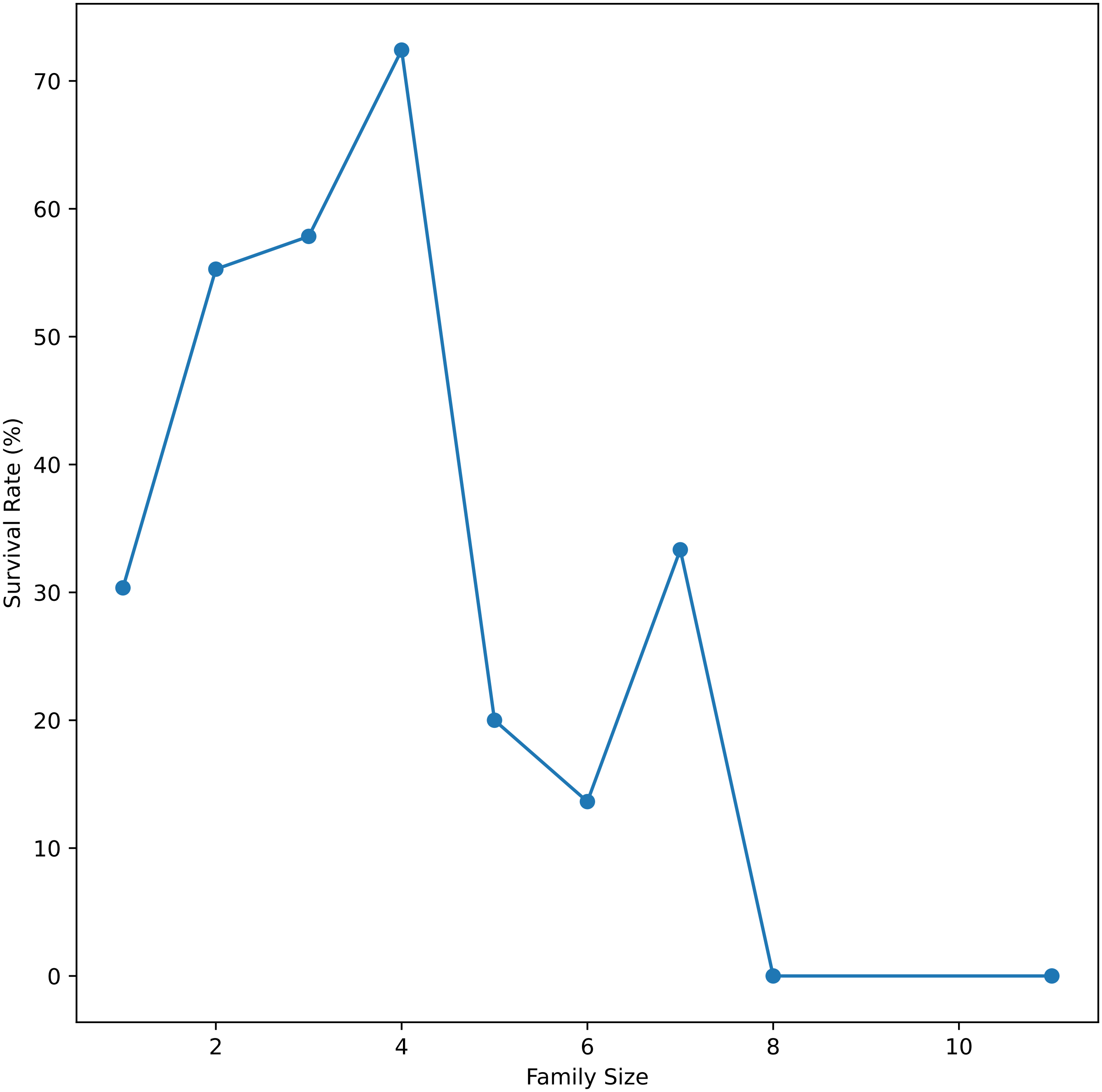
14. Correlation Heatmap



Observation

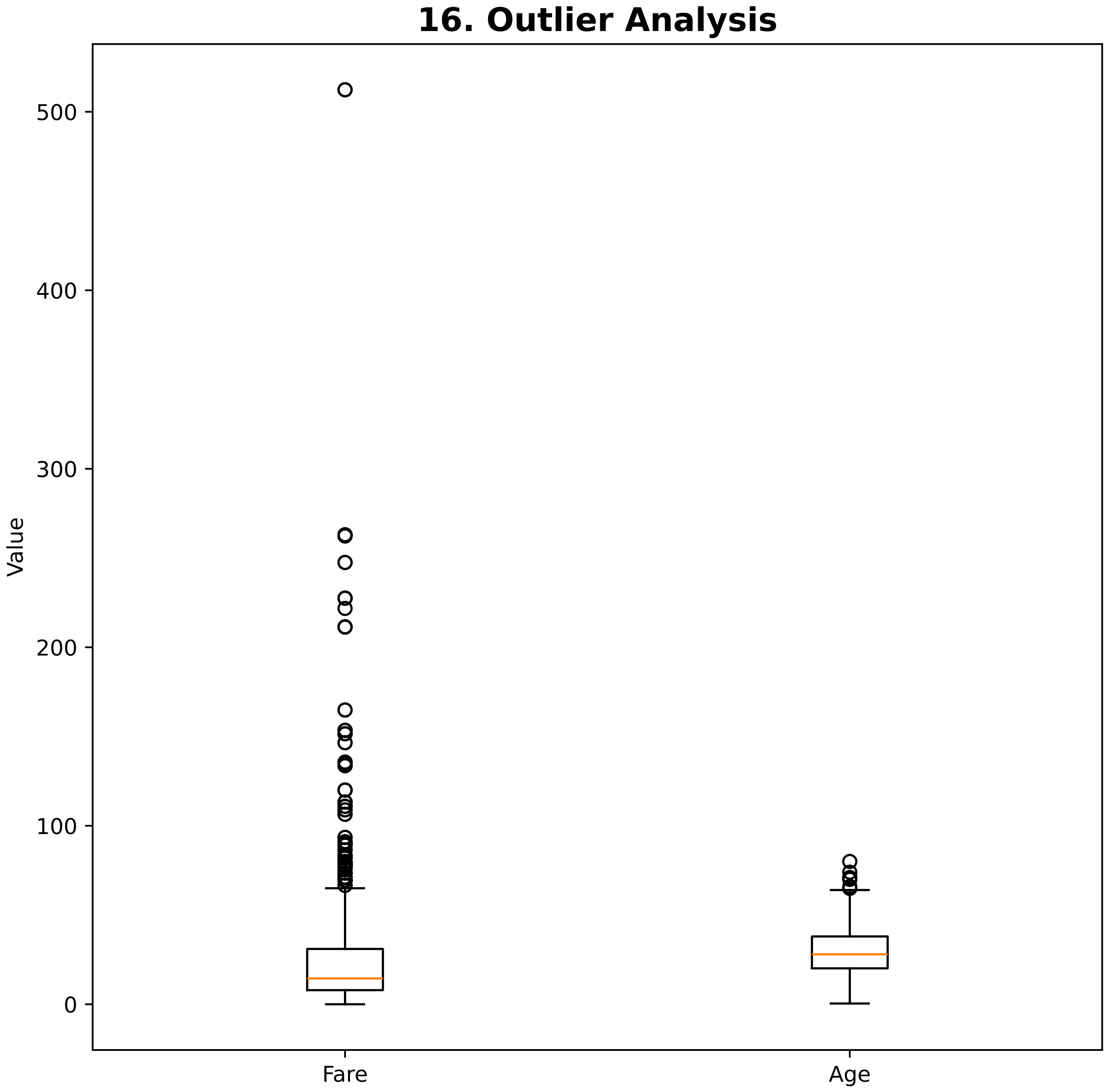
The heatmap shows numeric relationships. Survived is positively associated with Fare and negatively associated with Pclass. No pair of core numeric variables shows an extremely high correlation suggesting severe multicollinearity by itself.

15. Feature Engineering — Family Size



Observation

FamilySize = SibSp + Parch + 1. Survival patterns vary by family size, although very large families have fewer observations and should be interpreted cautiously.



Observation

Fare contains clear high-value outliers, consistent with its strong right skew. Outliers should be investigated rather than automatically removed.

17. Skewness Analysis

Fare

Skewness = 4.79. Fare is strongly right-skewed. A log1p transformation may be considered when appropriate.

Age

Skewness = 0.39. Age is less skewed than Fare, but missing values must be handled.

General approach

Inspect whether extreme values are genuine, then consider transformations, robust statistics or suitable models depending on the goal.

18. Key Findings and Conclusion

Key findings

1. 38.4% of passengers survived.
2. Female survival (74.2%) was much higher than male survival (18.9%).
3. First class had the highest survival rate and third class the lowest.
4. Fare is strongly right-skewed and contains high-value outliers.
5. Age has 177 missing values and Cabin has 687 missing values.
6. Sex + Pclass together provide a clearer survival pattern than either variable alone.
7. Correlation is useful for numeric relationships but does not prove causation.

Conclusion

The Titanic EDA reveals strong survival differences by sex and passenger class, along with important data-quality issues and skewed fare values. These findings provide a useful basis for further statistical analysis or predictive modeling.

19. Interview Questions — Prepared Answers

1. What is EDA and why is it important?

EDA examines data with statistics and visualizations to understand distributions, relationships, trends, anomalies and data-quality problems before modeling.

2. Which plots check correlation?

Correlation heatmaps summarize relationships among numeric variables; scatterplots examine relationships between two numeric variables.

3. How do you handle skewed data?

Inspect the cause and validity of extreme values, then consider transformations such as log1p, robust statistics or suitable models.

4. How do you detect multicollinearity?

Use a correlation matrix for an initial check and Variance Inflation Factor (VIF) for a formal predictor-level assessment.

5. Univariate, bivariate and multivariate?

Univariate analyzes one variable; bivariate examines two variables; multivariate examines three or more variables together.

6. Heatmap vs pairplot?

A heatmap displays values in a matrix, such as correlations. A pairplot shows pairwise distributions and relationships among selected variables.

7. How do you summarize insights?

State the strongest patterns, comparisons, anomalies and data-quality issues, tying each conclusion to statistics or visuals.